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DICTYOSTELIUM TRANSIENTLY EXPRESSES EIGHT DISTINCT G-PROTEIN α-SUBUNITS DURING ITS DEVELOPMENTAL PROGRAM

Lijun Wu and Peter N. Devreotes[1]

Department of Biological Chemistry
The Johns Hopkins University
School of Medicine
Baltimore, MD 21205

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Guanine nucleotide-binding proteins (G-proteins) play important roles in signal transduction and in controlling the developmental program of *Dictyostelium discoideum*. While five G-protein α-subunits have been identified in this organism, several lines of evidence suggest that additional G-proteins exist. Using the polymerase chain reaction technique, we have isolated three novel G-protein α-subunit genes. The predicted amino acid sequences of these PCR fragments are very similar to other *Dictyostelium* G-protein α-subunits in the conserved regions and share little homology outside of those regions. Northern blot analyses indicate that each of these genes has very distinct pattern of expression during development. The possible roles of these G-protein α-subunits are discussed. © 1991 Academic Press, Inc.

When deprived of nutrients, *Dictyostelium discoideum* amoebae cease growth and initiate a relatively simple and rapid developmental program. This program consists of the aggregation of 10^5 individual amoebae to form a multicellular mound that differentiates into a migrating slug and eventually culminates to form a fruiting body. This spontaneous process is organized by extracellular adenosine 3',5'- monophosphate (cAMP), which binds to surface receptors that mediate chemotaxis and oscillatory cell-cell signaling (1). The cAMP receptors contain seven transmembrane domains (2,3), a characteristic of G-protein-linked receptors.

G-proteins are membrane-associated heterotrimers composed of α, β, and γ subunits. In most systems, the α subunit determines the specificity of the interaction with receptors and effector molecules. Five G-protein α-subunit genes have been cloned from *Dictyostelium* (4-6). All of the genes encode the highly conserved domains characteristic of α-subunits. The time courses of expression of these genes have been characterized and some of them have been shown to play important roles in signal transduction and

[1] To whom correspondence should be addressed.

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in controlling the developmental program (5,7,8, Hadwiger and Firtel, personal communication). Gα2, for example, is required for the early aggregation process and most of the known cAMP receptor-mediated responses (7,8) and Gα4 plays a role in morphogenesis of the multicellular structure (Hadwiger and Firtel, personal communication). The functions of the other G-proteins are being investigated. However, several lines of evidence suggest that additional G-proteins exist. First, cellular responses involved with chemotaxis to folic acid are thought to be mediated by G-proteins coupling with cell surface receptors since folic acid binding is modulated by guanine nucleotides (9-12). Secondly, *Dictyostelium* adenylate cyclase is strongly activated by GTP and GTP-γ-S (13,14). The G-proteins identified previously are unlikely to fulfill these roles, since neither folic acid chemotaxis nor the activation of adenylate cyclase was affected in Gα1-, Gα2-, Gα3-, or Gα4- disrupted cells (5,8,15, Hadwiger and Firtel, personal communication), and Gα5 is expressed much later in the development (Hadwiger and Firtel, personal communication). Therefore, we decided to further explore the diversity of G-proteins in this organism. This report describes the discovery of three additional α-subunits.

MATERIALS AND METHODS

Polymerase chain reactions (PCRs). PCR was carried out using either *Dictyostelium* AX-3 genomic DNA or cDNA generated from RNA isolated from cells 3 hrs after nutrient depletion. Primers LW1 and LW4 correspond to the sense strand that encodes the amino acid sequence G(AT)GESGK and D(AV)(GA)GQR respectively; and primers LW2 and LW3 correspond to the antisense strand that encodes the amino acid sequence TCA(TV)DT and D(AV)(GA)GQR respectively. The sequences of the primers are:

LW1 5'CCGGATCCGG(AT)(AG)C(CAT)GG(AT)GA(GA)(AT)(GC)(CAT)GG(AT)AA;
LW2 5'CCGTCGACGT(AG)TC(ATG)(AG)(TC)(GAT)GC(GA)CA(GAT)GT;
LW3 5'CCGTCGAC(TG)(TC)TG(TA)CC(GAT)(CG)C(GAT)(GA)C(GA)TC;
LW4 5'CCGGATCCGA(CT)G(CT)(CTA)G(GC)(TAC)GG(TA)CA(AG)(AC)G.

The sequence of LW1 and LW4 contains a *Bam*HI restriction site and the sequence of LW2 and LW3 contains a *Sal*I restriction site to facilitate subcloning. Thirty amplification cycles consisting of 5 min at 94°C, 1.5 min at 55°C, and 3 min at 72°C were performed, followed by a final extension step (8 min at 72°C). The reaction products were purified by GeneClean (Bio101, La Jolla, CA), digested by *Bam*HI and *Sal*I, and subcloned to pBluescript KS/- (Stratagene, Burlingame, CA).

Southern Hybridization. Southern blot hybridization and colony hybridization were carried out as described by Maniatis *et al.* (16). DNA probes were made by the random priming method (17).

DNA sequence analysis. DNA sequencing reactions were performed using the Sequenase, version 2.0 kit according to the instructions provided by the manufacturer (United States Biochemical, Cleveland, OH). Double strand miniprep DNA for use as a sequencing template was prepared by the boiling lysis method (16). Clones were first grouped according to the pattern obtained from a single lane of DNA sequencing, and the complete DNA sequence was determined for representatives of each pattern.

Northern blot analysis. For the development time course of Gα gene expression, *D. discoideum* strain AX-3 cells grown in HL-5 medium were washed and plated on DB plates for development (18). Total RNA was isolated from cells harvested at specific times in development by the guanidine-HCl procedure (19), sized on a denaturing gel, and analyzed by RNA blot hybridization as described (16).

RESULTS AND DISCUSSION

Cloning of PCR fragments encoding for Gα6, Gα7, and Gα8 subunits. To isolate additional *Dictyostelium* α-subunit genes, we used short, degenerate primers in polymerase chain reactions on *Dictyostelium* genomic DNA or cDNA. The primers correspond to the amino acid sequences of the putative GTP-binding and GTPase regions and other highly conserved regions of known α-subunits. The PCR fragments were subcloned, and the clones containing Gα1 to Gα5 genes were eliminated by hybridization. The newly discovered α-subunits were identified by comparing the predicted amino acid sequence of each clone to the sequence of the known Gα proteins in the most conserved regions.

Three sets of primers were used in the PCR reactions. When primers LW1 and LW2, or LW2 and LW4 were used, only Gα1 to Gα5 clones and clones containing unknown sequences were obtained. These unknown sequences did not have any homology to the known Gα genes and they did not contain any major open reading frames, thus they are probably PCR artifacts. When primers LW1 and LW3 were used in PCR, about 1/3 of the clones analyzed fell into three novel α-subunit classes designated as Gα6, Gα7, and Gα8. The others contained unknown sequences and were not further studied. Gα6 clones were obtained only when using genomic DNA as a template, and Gα7 clones were obtained only when using cDNA as a template. Gα8 clones, however, were obtained when using either genomic DNA or cDNA as a template. Genomic Southern blot analyses showed that Gα6, Gα7, and Gα8 PCR fragments recognize distinct bands of *Dictyostelium* genomic DNA (data not shown). These results indicate that Gα6, Gα7, and Gα8 clones are amplified from contiguous segments of *Dictyostelium* DNA.

Sequence comparisons among *Dictyostelium* Gα subunits. The nucleotide sequences of Gα6, Gα7, and Gα8 PCR products were determined and a comparison of the predicted amino acid sequences of these PCR fragments to the previously characterized Gα subunits of *Dictyostelium* is shown in figure 1. Gα6, Gα7, and Gα8 are 25 - 31% identical to each other and are 32 - 41% identical to Gα1 and Gα2. These overall percentages are expected to be low since the amplified region is the least conserved stretch among all α-subunits.

```
          LW1
        ------->
        GA/TGESG
Gα1:    GAGESGKSTIAKQMKIIHLNGFNDE EKSSYKTIIYNNTVGSMRVLVNAAEELKIGISENNKEAA   SRISN   DLGDHFN   GV
Gα2:    GAGESGKSTISKQMKIIHQSGYSNE ERKEFKPIITRNILDNMRVLLDGMGRLGMTIDPSNSDAAVMIKELTSLQASIVTDCW   GE
Gα3:*         KSTVVKQMKILHHKGFSKE
Gα4:+         KSTIFKQMKIIQEDGGYSV
Gα5:+         KSTIFKQMKIIQDDGGFSI
Gα6:          KSTIFKQLKILQENGLWTTKELQEFSNTIYINIITQLQVLINAAESIGIQLQPENIQLS   KDYLELNHSSVTWTKEI GE
Gα7:          KSTVVKQMKILHLEGYSQE ERINQRQFVYRNIIEIAYSIIRGCGVLNLTIPSQFDSICSSIEEIYETKNYTNLDKNVLKGV
Gα8:          KSTVVKQLKSIYKIQVDDD ELHSYAVNIHKNTVLCMQVLLEAGETLGIELTDPETKKRAINVKSFQFEPEVKQMPVSI GL
Consensus:    KST  KQMKI    G     E       I  N      VL      L                                G

                                                                                        LW3
                                                                                    <---------
                                                                                    DA/VG/AGQR
Gα1:    LTAELAQDIKALWADPGIQNTFQRSSEF QLNDSAAYYFDSIDRISQPLYLPSENDVLRSRTKTTGIIETVFEIQNSTFRMVDVGGQR
Gα2:    LNEDQGKKIKALWTDPGVKQAMRRANEFSTLPDSAPYFFDSIDRMTSPVYIPTDQDILHTRVMTRGVHETNFEIGKIKFRLVDVGGQR
Gα3:
Gα4:
Gα5:
Gα6:    NSVR        LWNDSGIKSTFEKRDKEFQLNDSAEYFFNNLQRVTQDNYIPTPQDALRARVTTKGIIEADVTFDNINMKII
Gα7:    AE          LSKNESFINAANNSGSNFQLHSSSQYFLDDIARFSEEDYIPTDQDILYTRVASTSVSETRFSVRGIKFRMI
Gα8:    DIEE        LWRDEDIQEIWDRRSEYWFLDATPYYFEN IQRFLDDDFVPTEEDCIMTRVRTTGISVTEFDEGPVHFRVV
Consensus:          LW D                L  S  YF   I R     Y PT  D L  RV T G  ET F      FR
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Figure 1. Comparison of the deduced amino acid sequences of Gα6, Gα7, and Gα8 PCR products to that of the other *Dictyostelium* G-protein α-subunits. Amino acid sequences for Gα1 and Gα2 are from Pupillo *et al.* (4). *: The complete sequence of Gα3 is shown in Pupillo *et al.* (5). +: The cloning and sequencing of Gα4 and Gα5 are shown in Hadwiger *et al.* (6). A short segment of Gα3, Gα4, and Gα5 sequences are shown here to distinguish them from other G-protein α-subunits. The sequences of Gα7 and Gα8 were obtained from PCR fragments amplified from *Dictyostelium* cDNA and the sequence of Gα6 was obtained from a PCR fragment amplified from genomic DNA. An intron was found in the Gα6 clones and the coding sequence was derived based on the consensus exon-intron boundary sequences. The amino acid sequences chosen to design the PCR primers are shown on the top with the name and direction of each oligonucleotide. The consensus amino acid sequences were compiled from the sequences of each G-protein α-subunit shown. Amino acids in boldface are identical to the consensus sequences.

However, some amino acid residues are highly conserved among all the Gα subunits. A consensus amino acid sequence was compiled from the most conserved positions (occurring in more than 75% of the cases) in the amplified region within the primed sites. Each Gα sequence contains more than 82% of the consensus amino acids. Outside of these regions, the amino acid sequences show little homology. Based on this limited sequence information, the *Dictyostelium* G-proteins do not bear a direct relation to the α-subunit classes, such as Gs, Gi, Gq, etc., found in higher eucaryotes (20). Nor do the eight α-subunits fall into subtypes.

The expression of Gα6, Gα7 and Gα8 is developmentally regulated. RNA isolated from wild-type *Dictyostelium* (strain AX-3) cells at different stages of development was probed with Gα6, Gα7, and Gα8 PCR fragments (figure 2). Gα6 was expressed primarily in vegetative cells. The level of the Gα6 mRNA (about 1.8 kb) declined rapidly upon the initiation of development. Three hours after nutrient depletion (corresponding to very early aggregation), only a very weak signal was observed, and by 6 hr no band was

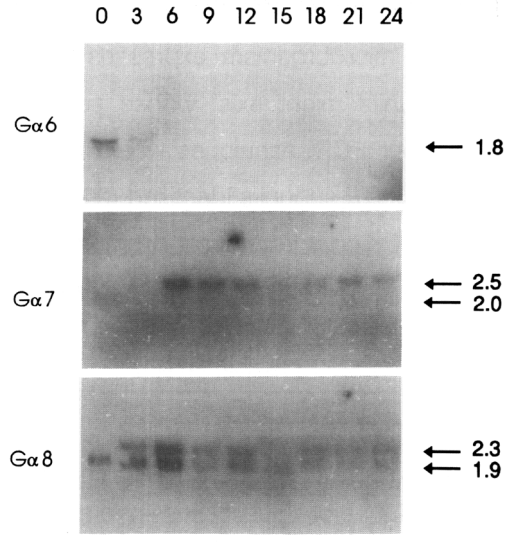

Figure 2. Developmental regulation of Gα6, Gα7, and Gα8 expression. Total RNA isolated from *Dictyostelium* wild-type (AX-3) cells at the times (hr) indicated was fractionated on a denaturing gel; blotted onto nitrocellulose filters; and probed with Gα6, Gα7, and Gα8 PCR fragments. Each lane contained 6 μg RNA. "0" indicates RNA isolated from vegetative cells. During this developmental time course, visible aggregates were observed at 5-6 hr, mounds were formed at 9-10 hr, and fruiting bodies were found at 24 hr. Numbers on the right indicate the size of the RNAs in kb.

detected. Gα7 hybridized to two mRNA species (2.0 kb and 2.5 kb respectively) at different stages of development. A minor 2.0-kb band was detected in growing cells and was absent in starved cells. The level of the major Gα7 RNA (2.5 kb) peaked in late aggregation and early mound stages (6 to 9 hrs), and declined thereafter. Gα8 also hybridized to two mRNAs (1.9 kb and 2.3 kb respectively). The 2.3 kb species was not expressed or expressed at very low levels in vegetative cells. After the initiation of development, levels of this RNA species were induced and both species reached a maximal level during aggregation (between 3 and 6 hrs). The levels of the Gα8 RNAs declined in the late development.

The distinct expression pattern of Gα6, Gα7, and Gα8 genes suggest that these G proteins may be involved in different signal transduction pathways. Gα6 is expressed primarily in vegetative cells and thus it might be involved in the folic acid chemotaxis signal transduction pathway. The time course of Gα8 correlates with that of the adenylate cyclase activity in *Dictyostelium* (21,22) and is a potential analogue for the Gs-like G-proteins which stimulate adenylate cyclase in mammals. To further study these novel Gα genes, experiments are in progress to screen for genomic and cDNA clones, to determine the complete nucleotide sequences, and to make gene-disruption mutants to investigate their functions.

In conclusion, the discovery of three additional Gα genes in *Dictyostelium* suggests that this primitive multicellular microorganism expresses at least eight G-protein α-subunits. More novel α-subunits might exist whose sequence is divergent in the conserved regions such that our PCR techniques would not have detected them. In mammalian system, there are 17 Gα subunits identified (20). The remarkable degree of similarity in the mechanisms of signal transduction among *Dictyostelium* and higher eukaryotes makes this biochemically and genetically accessible organism an attractive model system to study G-protein function.

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NOTE

PCR fragments containing sequences very similar to that of the Gα6 clones described here were also obtained independently in another lab (R. Firtel, personal communication). The two clones were different for three amino acid residues.

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